MAJOR PROJECT 3

College Admission Prediction
Data Cleaning, Exploratory Data Analysis & Accuracy Improvement

Project Report

This project develops a machine-learning workflow for predicting college admission status. The work covers dataset inspection, cleaning, missing-value treatment, duplicate checking, outlier analysis, exploratory data analysis, visualization, feature engineering, baseline modeling, advanced modeling, and performance comparison.

1. Dataset Overview

The project uses the College Admission dataset. The notebook loads the raw CSV, examines its structure, data types, descriptive statistics, missing values, duplicate rows, unique values, and target distribution. The cleaned dataset is exported as **College_Admission_cleaned.csv**.

2. Data Cleaning

The cleaning workflow removes completely empty rows and exact duplicate rows, strips unnecessary whitespace from column names and categorical values, standardizes categorical text to lowercase, and removes **student_id** from predictive modeling because it is an identifier rather than a meaningful predictive feature. Numerical missing values are treated using median imputation and categorical missing values using the most frequent category.

3. Outlier Detection

Numerical variables are checked using the Interquartile Range (IQR) method. For every numerical feature, Q1, Q3, IQR, lower bound, upper bound, and the number of observations outside the bounds are calculated. Boxplots are used to visualize these observations. Outliers are reported rather than automatically deleted, because extreme applicant characteristics can represent legitimate observations.

4. Exploratory Data Analysis

The notebook contains extensive visual analysis, including target-class distribution, age distribution, entrance-score distribution, board-percentage distribution, extracurricular-score distribution, admission-probability distribution, categorical count plots, admission status by category, boxplots, scatter plots, pairplot, and a numerical correlation heatmap. These plots help identify distributions, relationships, class patterns, and potential influential variables.

5. Feature Engineering

Additional features are created from existing academic variables. These include an academic average, score-to-board gap, entrance-board interaction, extracurricular-to-board ratio, and squared age. These transformations are intended to help the improved model capture nonlinear relationships and interactions.

6. Baseline Model

Logistic Regression is used as the baseline classifier. Categorical variables are encoded using one-hot encoding, while numerical features are imputed and standardized through a preprocessing pipeline. The target

is encoded as 0 for rejected and 1 for admitted.

7. Improved Model

The enhanced experiment uses feature engineering followed by HistGradientBoostingClassifier. This model can capture nonlinear patterns and interactions that a linear baseline may not represent effectively. The notebook calculates accuracy, precision, recall, F1-score, and ROC-AUC for the improved model.

8. Model Evaluation & Comparison

The notebook generates a side-by-side comparison of the baseline and improved models. It also calculates the relative accuracy improvement using: **((Improved Accuracy** – **Baseline Accuracy) / Baseline Accuracy) × 100**.

Evaluation Metric	Baseline	Improved
Accuracy	Calculated in notebook	Calculated in notebook
Precision	Calculated in notebook	Calculated in notebook
Recall	Calculated in notebook	Calculated in notebook
F1 Score	Calculated in notebook	Calculated in notebook
ROC-AUC	Calculated in notebook	Calculated in notebook

The final notebook also produces a metrics comparison graph, improved-model confusion matrix, ROC curve, and classification reports.

9. Key Visual Outputs

The completed notebook includes more than 20 visual outputs covering distributions, categorical frequencies, target relationships, numerical boxplots, correlation, pairwise relationships, model comparison, confusion matrix, and ROC analysis.

10. Conclusion

The project demonstrates a complete end-to-end machine-learning workflow for college admission prediction. Data quality is assessed before modeling, categorical and numerical variables are prepared appropriately, outliers are investigated rather than blindly removed, and extensive EDA is used to understand the dataset. A Logistic Regression baseline is then compared with a feature-engineered HistGradientBoosting model. The final notebook calculates all evaluation metrics and the percentage improvement directly from the test data, making the result reproducible.

11. Deliverables

• Complete Jupyter Notebook (.ipynb)
• Cleaned dataset (.csv)
• Final project report (.pdf)
• Graphs, metrics, confusion matrix and ROC analysis generated within the notebook